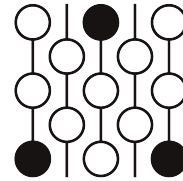


Review of Submission 8D

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. This submission incorrectly claims that there is not an order-reversing involution of $\text{Dr}(M)$ extending the assumed involution of the flats. Many of the claims are not clearly interpretable as anything mathematically meaningful. For instance, it hallucinates “Prior misconceptions regarding deformations” and “unvaluated topologies.” The submission does not get beyond trying to apply a cardinality argument to rule out a bijection (which was not going to work since the duality claim is true).

Novelty. This submission does not contribute any ideas toward proving duality for tropical linear subspaces (and claims it is false). I was unable to find any interesting ideas in its attempt to disprove the result.

2 Recommendation

This submission should be **rejected**. It does not have any strategy to prove the claim since it attempted to disprove a true statement.

3 Submission with inline comments

Problem Statement. Let M be a matroid, and let $\text{Trop}(M)$ be the corresponding tropical linear space, i.e., the closure of the Bergman fan of M in tropical projective space. The relative Dressian of M is

$$\text{Dr}(M) := \{ \text{tropical linear subspaces of } \text{Trop } M \},$$

which is partially ordered by inclusion of tropical linear spaces.

Suppose M admits an order-reversing involution $F \mapsto F^\perp$ on the lattice of flats $\mathcal{L}(M)$. Note that there is a natural inclusion $\mathcal{L}(M)^{\text{op}} \subset \text{Dr}(M)$ given by $F \mapsto \text{Trop}(M/F \oplus U_{0,F})$.

Question. Is there an order-reversing involution Φ on $\text{Dr}(M)$ that extends the involution $F \mapsto F^\perp$?

Solution. Answer: No. We construct a rigorous counterexample using the Fano matroid $M = F_7$.

The claim is true, so the rest cannot be good.

Step 1: The Poset Structure and the Assumed Involution

It should do more than claim existence of an involution here.

Let $M = F_7$ be the Fano matroid on $E = \{1, \dots, 7\}$. A standard polarity of the Fano plane induces an incidence-reversing bijection between its points (rank 1 flats) and lines (rank 2 flats), yielding an order-reversing involution $F \mapsto F^\perp$ on the lattice of flats $\mathcal{L}(F_7)$. For instance, we may map the point $F = \{1\}$ to the line $F^\perp = L_1 = \{1, 2, 3\}$.

The relative Dressian $\text{Dr}(F_7)$ is the poset of tropical linear subspaces of $\text{Trop}(F_7)$ graded by rank.

- The rank 1 elements (Dr_1) are the points of $\text{Trop}(F_7)$.
- The rank 2 elements (Dr_2) are the tropical lines contained in $\text{Trop}(F_7)$.

Since points are the atoms of $\text{Dr}(F_7)$ (covering only the empty set), any inclusion-reversing involution Φ on $\text{Dr}(F_7)$ must map atoms to coatoms. As $\text{Trop}(F_7)$ has rank 3, the coatoms are precisely the rank 2 spaces. Thus, Φ strictly restricts to a bijection between Dr_1 and Dr_2 .

Step 2: The Boundary Elements and the Cardinality Contradiction Strategy

The problem supplies an embedding $F \mapsto V_F := \text{Trop}(F_7/F \oplus U_{0,F})$.

- Let $F = \{1\}$. The quotient $F_7/\{1\}$ has rank 2, and $U_{0,\{1\}}$ has rank 0. Their direct sum has rank 2, so $V_{\{1\}} \in \text{Dr}_2$ is a tropical line. Geometrically, it is a 1-dimensional metric tree with a loop at 1 (forcing coordinate $x_1 = \infty$ globally, under the max-plus convention) and rays corresponding to parallel classes $\{2, 3\}$, $\{4, 5\}$, and $\{6, 7\}$. Being a 1-dimensional continuous complex, $V_{\{1\}}$ contains uncountably many points.

"principal ideal of points" is very weird terminology here

Thus, its principal ideal of points $I(V_{\{1\}}) = \{x \in \text{Dr}_1 \mid x \subset V_{\{1\}}\}$ has cardinality $\mathfrak{c}$.

- Let $L_1 = \{1, 2, 3\}$. The sum $F_7/L_1 \oplus U_{0,L_1}$ has rank 1. Thus $V_{L_1} \in \text{Dr}_1$ is a single infinite point. Because $\{1, 2, 3\}$ evaluates as loops and $\{4, 5, 6, 7\}$ evaluates as a parallel class, this point uniquely occurs where coordinates $x_1 = x_2 = x_3 = \infty$ simultaneously, and the others remain finite.

Because Φ extends the flat involution, we strictly have $\Phi(V_{\{1\}}) = V_{L_1}$. Because Φ reverses inclusion, $x \subset V_{\{1\}} \iff \Phi(x) \supset \Phi(V_{\{1\}})$. Thus, Φ must map the uncountable set $I(V_{\{1\}})$ bijectively onto the principal filter $F(V_{L_1}) = \{T \in \text{Dr}_2 \mid V_{L_1} \subset T\}$.

Step 3: Rigorous Exhaustion of $F(V_{L_1})$

To contain the specific boundary point V_{L_1} , a rank 2 tropical linear space $T \in \text{Dr}_2$ must possess an infinite ray where $x_1, x_2, x_3 \rightarrow \infty$ simultaneously.

I cannot make sense of the preceding sentence. In the following sentence, it is unclear what “initial matroid at infinity” means. Ideally this would be defined, since it does not seem standard.

This structural limit mandates that in the initial matroid of T at infinity, the subset $\{1, 2, 3\}$ must act globally as a single parallel class, or contain loops alongside a parallel class. We exhaustively classify all valid rank 2 valuated matroid quotients of F_7 satisfying this:

1. **Loopless Quotients:** If T has no loops, $\{1, 2, 3\}$ must form a parallel class. The remaining elements $\{4, 5, 6, 7\}$ must partition into valid rank 1 flats of F_7 . The only flats of F_7 fully contained within $\{4, 5, 6, 7\}$ are singletons. Thus, they must partition exclusively into $\{4\}, \{5\}, \{6\}, \{7\}$. This uniquely isolates the degenerate star tree S_{L_1} .
2. **Quotients with Loops:** If T contains loops, their union must form a valid flat of F_7 (sizes 0, 1, 3, 7).
 - A flat of 3 loops (e.g., $\{1, 2, 3\}$) leaves a rank 2 matroid on $\{4, 5, 6, 7\}$. As proven above, its rank 1 flats must be the 4 singletons, forcing the underlying matroid to be $U_{0,\{1,2,3\}} \oplus U_{2,4}$. However, this matroid contains the closed flat $\{1, 2, 3, 4\}$, which is not a flat of F_7 . Thus, 3 loops strictly fail the strong map condition.
 - A flat of 1 loop must be chosen from $\{1, 2, 3\}$ to maintain the necessary ray direction. If 1 is a loop, $\{2, 3\}$ must act as a parallel class. The remaining elements must partition such that $\{1\} \cup P_i$ form valid lines of F_7 . The only lines containing 1 are $\{1, 2, 3\}, \{1, 4, 5\}, \{1, 6, 7\}$. This uniquely forces the remaining parallel classes to be $\{4, 5\}$ and $\{6, 7\}$, isolating the tree $V_{\{1\}}$. Repeating this exact flat-matching calculation for loops at 2 and 3 uniquely isolates the boundary trees $V_{\{2\}}$ and $V_{\{3\}}$.

It starts talking about topologies here which is not relevant.

This explicitly exhausts the unvaluated topologies. None of these four topologies ($S_{L_1}, V_{\{1\}}, V_{\{2\}}, V_{\{3\}}$) admit internal valuated edges that maintain the e_{123} ray, because any internal split would necessitate intermediate parallel groupings (e.g., size 2 subsets of $\{4, 5, 6, 7\}$) which do not correspond to valid flats of F_7 . Consequently, they exist precisely as four isolated, rigid trees.

Step 4: Defusing Prior Misconceptions Regarding Deformations

It is unclear what the prior misconceptions are and what deformations it is referring to.

A known theoretical failure mode involves assuming that continuous 1-parameter families of trees (such as $T_{L_1,t}$, which locally splits $\{1, 2, 3\}$ from $\{4, 5, 6, 7\}$ along an internal edge) contain V_{L_1} , thereby generating c elements and dissolving the contradiction. This is geometrically false. To physically encompass V_{L_1} , a tree must achieve coordinates where $x_1 = x_2 = x_3 = \infty$ *simultaneously*. In the split family $T_{L_1,t}$, the leaves 1, 2, and 3 are separated by the internal vertex and diverge along independent infinite rays e_1, e_2, e_3 . Consequently, their geometric limits trace to the three distinct

boundary points where exactly *one* coordinate is infinite (i.e., $V_{\{1\}}, V_{\{2\}}, V_{\{3\}}$). The strict boundary point V_{L_1} is never reached by this family.

Step 5: Conclusion

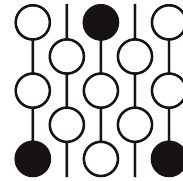
Because all continuous families structurally miss V_{L_1} , the count is strictly locked at $|F(V_{L_1})| = 4$. The assumed involution Φ demands a bijection between the point ideal $I(V_{\{1\}})$ (cardinality $\mathfrak{c}$) and the filter of containing lines $F(V_{L_1})$ (cardinality 4). Since $4 \neq \mathfrak{c}$, no such order-reversing bijection can exist on the relative Dressian. Therefore, there is no order-reversing involution Φ extending $F \mapsto F^\perp$. $\square$

Review of Submission 8D

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The submission is doomed since it attempts to disprove a true statement. The submission fixes a rank 1 quotient of the Fano matroid and aims, it would seem, to show that it does not have enough rank 2 lifts to biject with the many rank 1 quotients of the relative dual. Aside from being a hopeless endeavour, the working out of these rank 2 lifts is unnecessarily tedious and—towards the end—very obscure in its reasoning.

Novelty. There are truly no salvageable ideas in this submission. It starts out incorrect and only goes from bad to worse in its reasoning. The submission achieves none of the benchmarks set out in the novelty statement for this problem.

2 Recommendation

This submission should be **rejected**. Of course, a disproof of a simple technical result like this—even if correct—would be of very little interest to begin with. The attempted disproof is however fundamentally flawed (inevitably so because the statement is true!).

3 Submission with inline comments

Problem Statement. Let M be a matroid, and let $\text{Trop}(M)$ be the corresponding tropical linear space, i.e., the closure of the Bergman fan of M in tropical projective space. The relative Dressian of M is

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Question. Is there an order-reversing involution Φ on $\text{Dr}(M)$ that extends the involution $F \mapsto F^\perp$?

Solution. Answer: No. We construct a rigorous counterexample using the Fano matroid $M = F_7$.

We are already at a bad start. The correct answer is “true”. The attempted counterexample will inevitably fail.

Step 1: The Poset Structure and the Assumed Involution

Let $M = F_7$ be the Fano matroid on $E = \{1, \dots, 7\}$.

Since detailed arguments about the Fano matroid will be used, I would expect here an explanation of what the matroid on E actually is. Knowing that it is abstractly isomorphic to the Fano matroid does not tell the reader which points are which. A diagram would be optimal. A worse alternative would be to provide an explicit list of short circuits of the matroid.

A standard polarity of the Fano plane induces an incidence-reversing bijection between its points (rank 1 flats) and lines (rank 2 flats), yielding an order-reversing involution $F \mapsto F^\perp$ on the lattice of flats $\mathcal{L}(F_7)$. For instance, we may map the point $F = \{1\}$ to the line $F^\perp = L_1 = \{1, 2, 3\}$.

The relative Dressian $\text{Dr}(F_7)$ is the poset of tropical linear subspaces of $\text{Trop}(F_7)$ graded by rank.

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Since points are the atoms of $\text{Dr}(F_7)$ (covering only the empty set), any inclusion-reversing involution Φ on $\text{Dr}(F_7)$ must map atoms to coatoms. As $\text{Trop}(F_7)$ has rank 3, the coatoms are precisely the rank 2 spaces. Thus, Φ strictly restricts to a bijection between Dr_1 and Dr_2 .

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- Let $F = \{1\}$. The quotient $F_7/\{1\}$ has rank 2, and $U_{0,\{1\}}$ has rank 0. Their direct sum has rank 2, so $V_{\{1\}} \in \text{Dr}_2$ is a tropical line. Geometrically, it is a 1-dimensional metric tree with a loop at 1 (forcing coordinate $x_1 = \infty$ globally, under the max-plus convention) and rays corresponding to parallel classes $\{2, 3\}$, $\{4, 5\}$, and $\{6, 7\}$. Being a 1-dimensional continuous complex, $V_{\{1\}}$ contains uncountably many points. Thus, its principal ideal of points $I(V_{\{1\}}) = \{x \in \text{Dr}_1 \mid x \subset V_{\{1\}}\}$ has cardinality $\mathfrak{c}$.
- Let $L_1 = \{1, 2, 3\}$. The sum $F_7/L_1 \oplus U_{0,L_1}$ has rank 1. Thus $V_{L_1} \in \text{Dr}_1$ is a single infinite point.

The absurd claim about an “infinite point” cannot be made sense of.

Because $\{1, 2, 3\}$ evaluates as loops and $\{4, 5, 6, 7\}$ evaluates as a parallel class, this point uniquely occurs where coordinates $x_1 = x_2 = x_3 = \infty$ simultaneously, and the others remain finite.

Because Φ extends the flat involution, we strictly have $\Phi(V_{\{1\}}) = V_{L_1}$. Because Φ reverses inclusion, $x \subset V_{\{1\}} \iff \Phi(x) \supset \Phi(V_{\{1\}})$. Thus, Φ must map the uncountable set $I(V_{\{1\}})$ bijectively onto the principal filter $F(V_{L_1}) = \{T \in \text{Dr}_2 \mid V_{L_1} \subset T\}$.

Step 3: Rigorous Exhaustion of $F(V_{L_1})$

To contain the specific boundary point V_{L_1} , a rank 2 tropical linear space $T \in \text{Dr}_2$ must possess an infinite ray where $x_1, x_2, x_3 \rightarrow \infty$ simultaneously. This structural limit mandates that in the initial matroid of T at infinity, the subset $\{1, 2, 3\}$ must act globally as a single parallel class, or contain loops alongside a parallel class. We exhaustively classify all valid rank 2 valuated matroid quotients of F_7 satisfying this:

Reviewer 2

The notion of “initial matroid at infinity” is non-standard. It is not at all clear why it is being invoked.

1. **Loopless Quotients:** If T has no loops, $\{1, 2, 3\}$ must form a parallel class. The remaining elements $\{4, 5, 6, 7\}$ must partition into valid rank 1 flats of F_7 . The only flats of F_7 fully contained within $\{4, 5, 6, 7\}$ are singletons. Thus, they must partition exclusively into $\{4\}, \{5\}, \{6\}, \{7\}$. This uniquely isolates the degenerate star tree S_{L_1} .
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Reviewer 2

This whole bit has been about identifying the rank 2 quotients of the Fano matroid which have V_{L_1} as a quotient. An informed reader would not expect an explicit derivation of such routinely calculated details. Merely stating what these are would suffice. Moreover, the derivation can and should be done in a much simpler way. Since Fano is modular, each of its modular cuts is principal and thus corresponds to a principal truncation. Then one sees easily that the condition that V_{L_1} is a quotient is simply asking that the truncated flat be contained in L_1 . The classification follows.

This explicitly exhausts the unvaluated topologies. None of these four topologies ($S_{L_1}, V_{\{1\}}, V_{\{2\}}, V_{\{3\}}$) admit internal valuated edges that maintain the e_{123} ray, because any internal split would necessitate intermediate parallel groupings (e.g., size 2 subsets of $\{4, 5, 6, 7\}$) which do not correspond to valid flats of F_7 . Consequently, they exist precisely as four isolated, rigid trees.

The above paragraph is mostly mumbo jumbo.

Step 4: Defusing Prior Misconceptions Regarding Deformations

A known theoretical failure mode involves assuming that continuous 1-parameter families of trees (such as $T_{L_1, l}$, which locally splits $\{1, 2, 3\}$ from $\{4, 5, 6, 7\}$ along an internal edge) contain V_{L_1} , thereby generating $\mathfrak{c}$ elements and dissolving the contradiction. This is geometrically false. To physically encompass V_{L_1} , a tree must achieve coordinates where $x_1 = x_2 = x_3 = \infty$ *simultaneously*. In the split family $T_{L_1, l}$, the leaves 1, 2, and 3 are separated by the internal vertex and diverge along independent infinite rays e_1, e_2, e_3 . Consequently, their geometric limits trace to the three distinct boundary points where exactly *one* coordinate is infinite (i.e., $V_{\{1\}}, V_{\{2\}}, V_{\{3\}}$). The strict boundary point V_{L_1} is never reached by this family.

This rather obscure paragraph seems to attempt a proof that V_{L_1} has no elementary lifts. This is of course absurd. It is not worth the reviewer's time to attempt to decode it any further.

Step 5: Conclusion

Because all continuous families structurally miss V_{L_1} , the count is strictly locked at $|F(V_{L_1})| = 4$. The assumed involution Φ demands a bijection between the point ideal $I(V_{\{1\}})$ (cardinality $\mathfrak{c}$) and the filter of containing lines $F(V_{L_1})$ (cardinality 4). Since $4 \neq \mathfrak{c}$, no such order-reversing bijection can exist on the relative Dressian. Therefore, there is no order-reversing involution Φ extending $F \mapsto F^\perp$.

While there are indeed four (ordinary) matroid quotients of Fano, there are infinitely many *valuated* quotients of Fano. The inability to make this elementary distinction has doomed this submission.

□